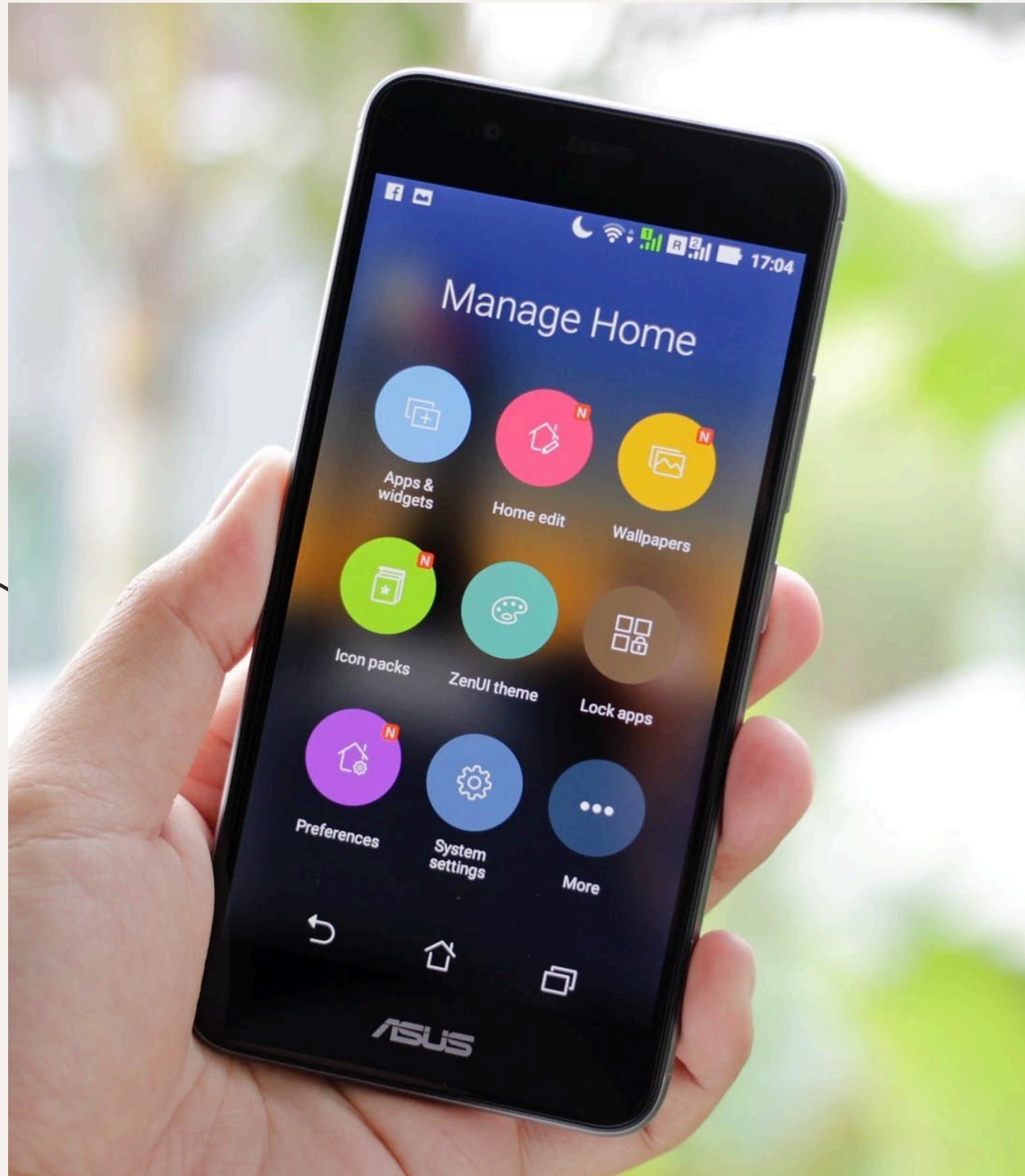


Project Title :- Mobile Price Prediction Model

Name :- AKSHAT JAIN

Date :- 14 November, 2024





Objective :-

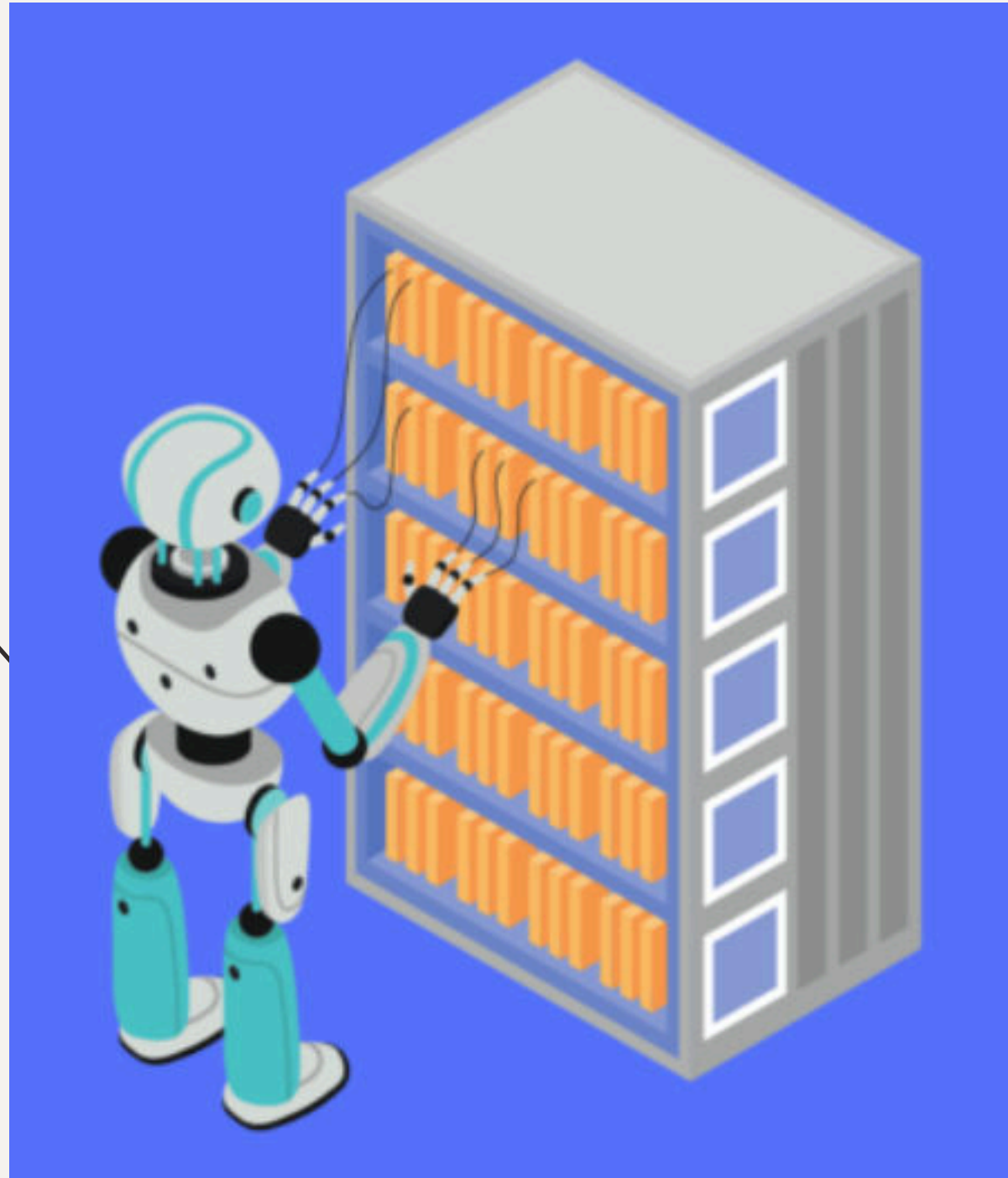
Develop a linear regression model to predict the price of mobile phones based on various features such as brand, name, cameras and market trends. The aim is to create an accurate pricing model that can estimate mobile phone prices effectively

Machine Learning Approach

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that enables systems to learn from data and improve performance over time without explicit programming.

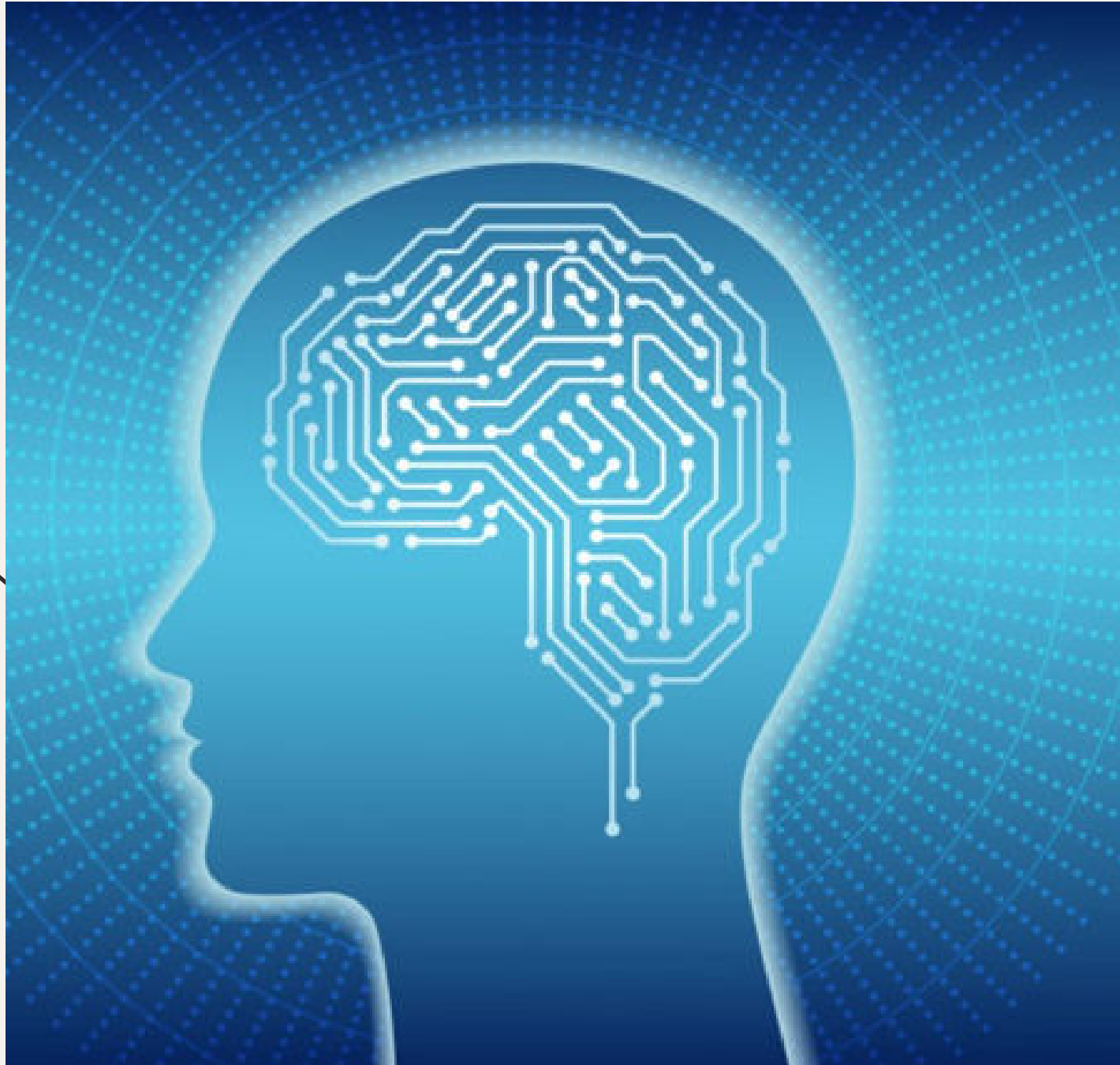
- 1)** ML models learn patterns and insights from historical data and make predictions or decisions based on new data.
- 2)** ML models can be trained using different approaches, depending on the availability of labeled data and the problem type.

Supervised Learning



- 1)** Supervised Learning involves training models on labeled data to predict outcomes for new, unseen data.
- 2)** It is commonly used for tasks like classification and regression (e.g., predicting mobile device failure).

Unsupervised Learning



- 1)** Unsupervised Learning works with unlabeled data to identify patterns or groupings in the data.
- 2)** It is typically used for clustering and anomaly detection (e.g., segmenting users based on app usage).

Reinforcement Learning



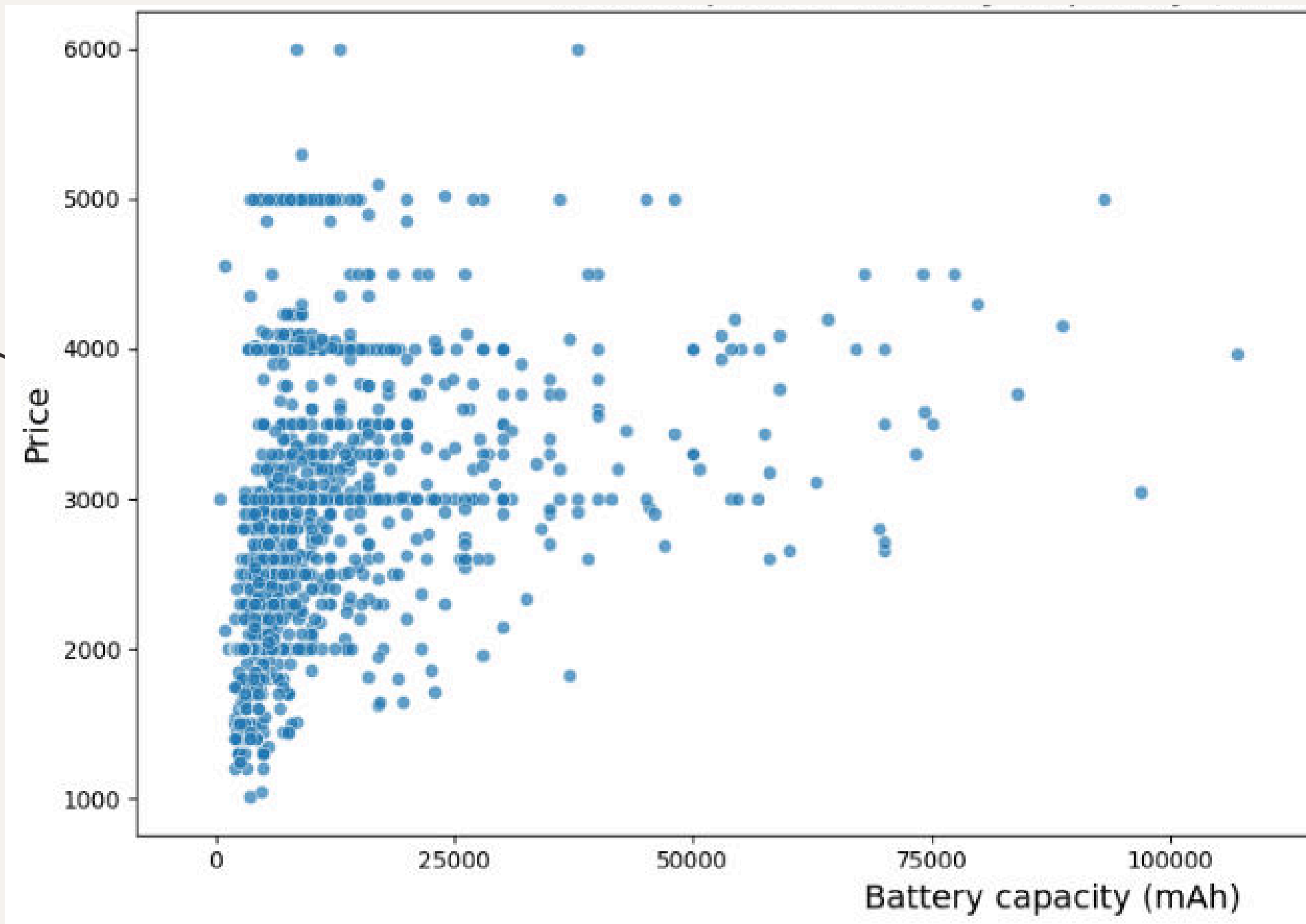
- 1) Reinforcement Learning teaches models to make decisions by receiving rewards or penalties based on their actions.
- 2) It is applied in dynamic environments, such as optimizing mobile battery usage or adaptive systems.



Exploratory Data Analysis

Identifying the right **data sources** is crucial for a successful machine learning project. We will discuss various types of **data**, including structured and unstructured, and how to evaluate their **quality** and **relevance** for your specific objectives.

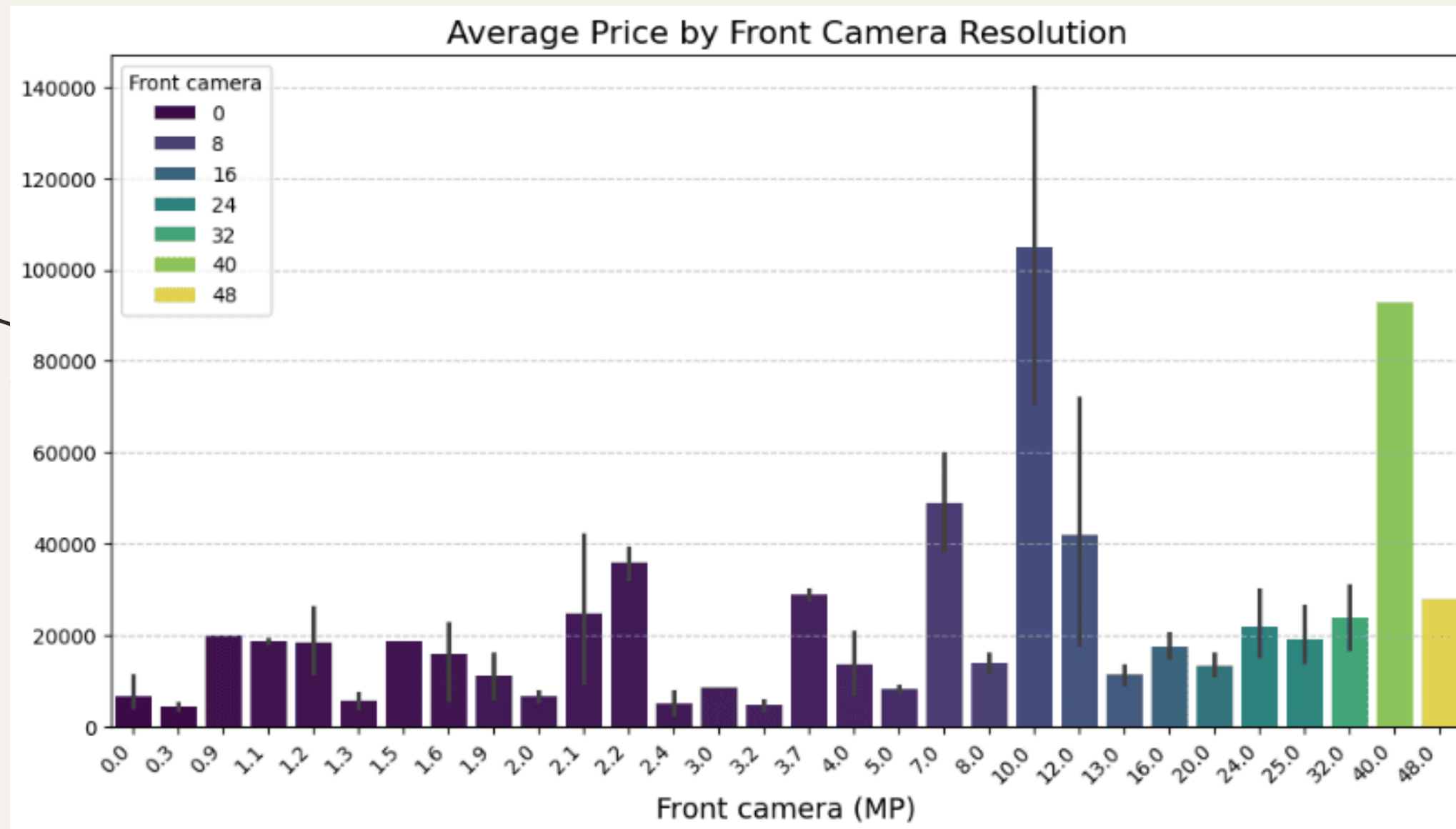
BATTERY CAPACITY AND PRICING



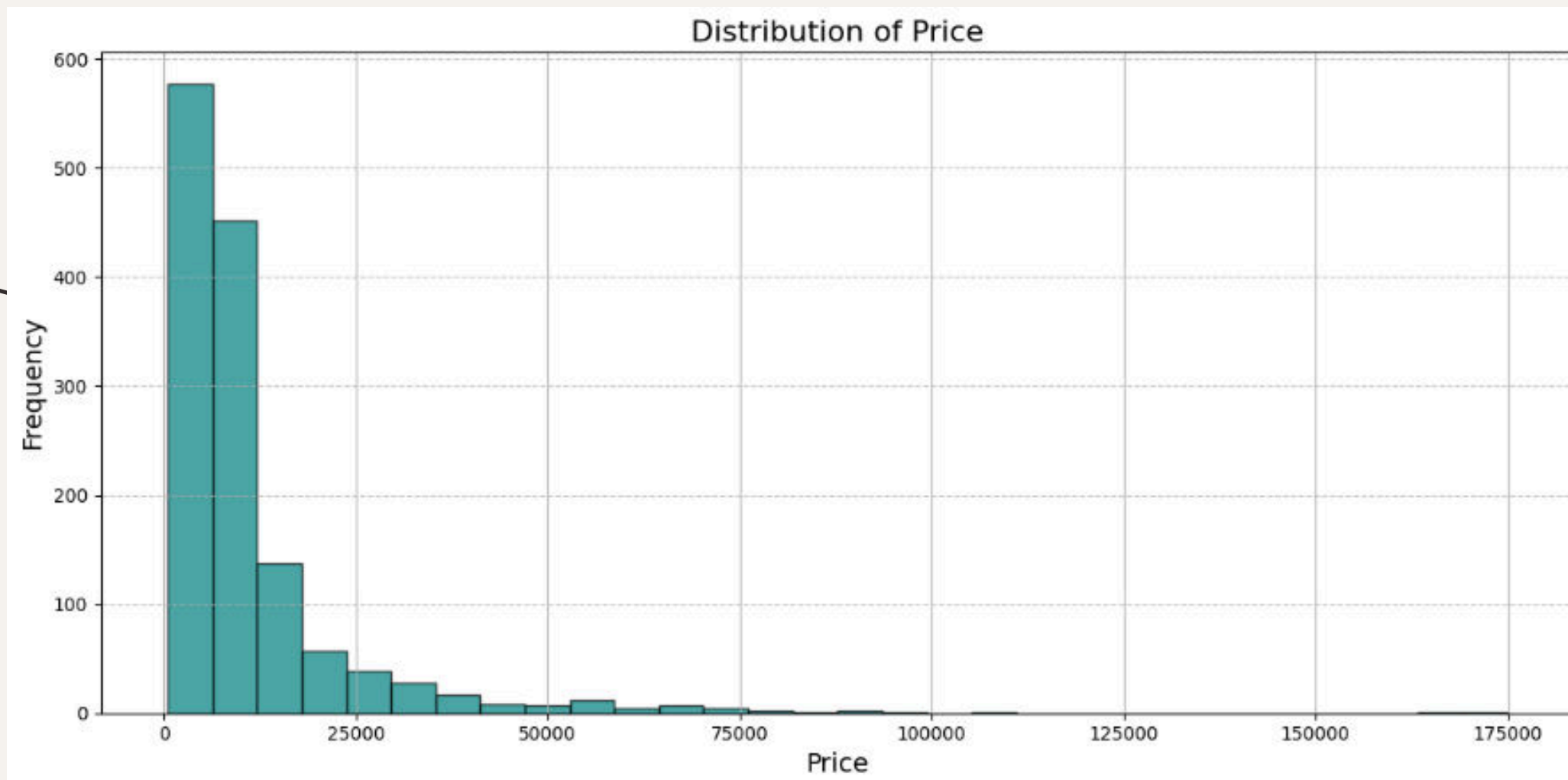
Phones with larger **battery** capacities tend to be priced **higher**. This is because extended battery life, which reduces the need for frequent **recharging**, adds value and justifies a higher price point.

CAMERA AND PRICING

The price of a Phone isn't always directly related to camera. For instance, mid-range phones might offer better camera for a better viewing experience, while high-end phone often prioritize a more compact design, which doesn't always mean a better camera.



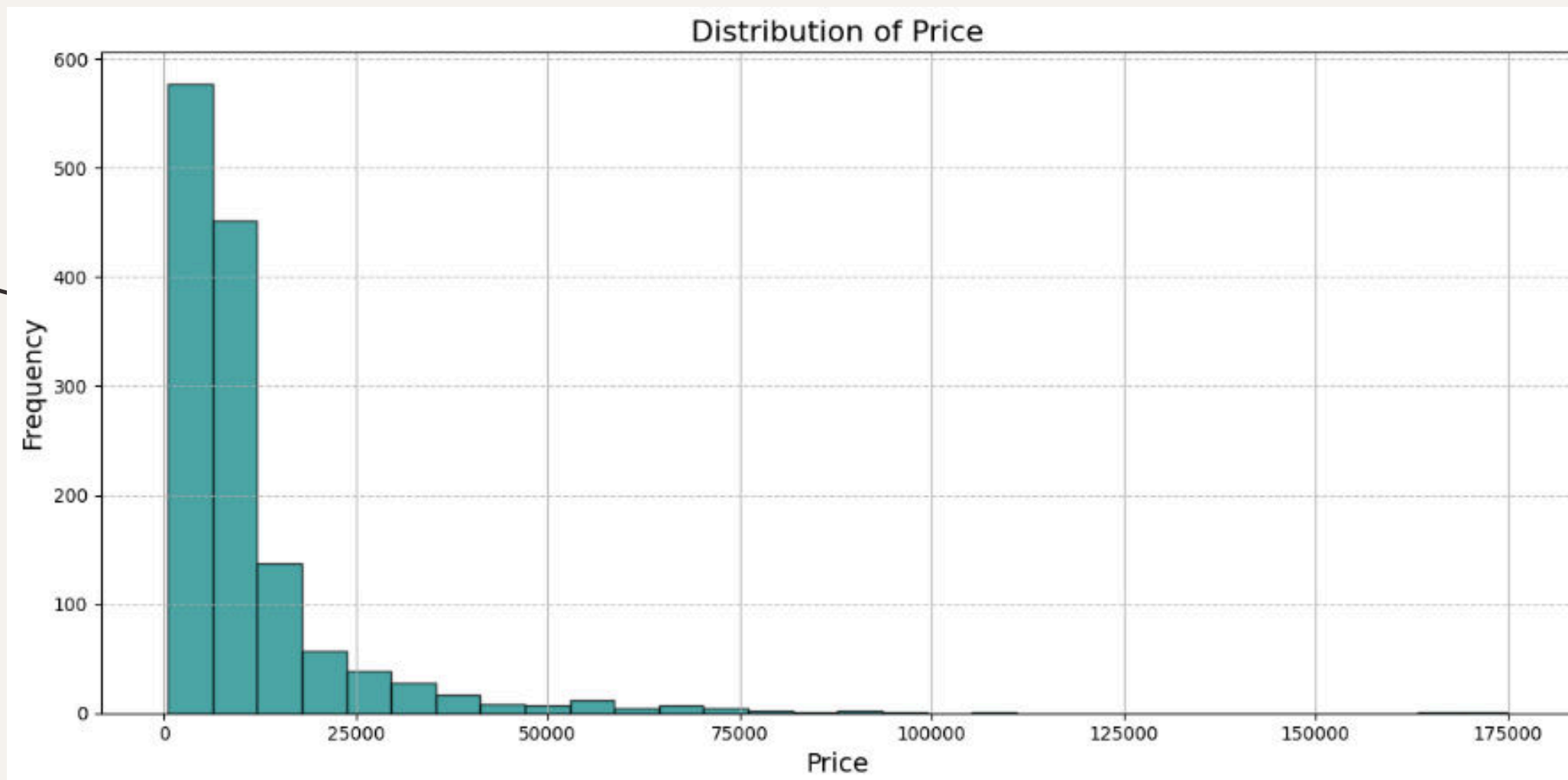
DISTRIBUTION OF PRICE



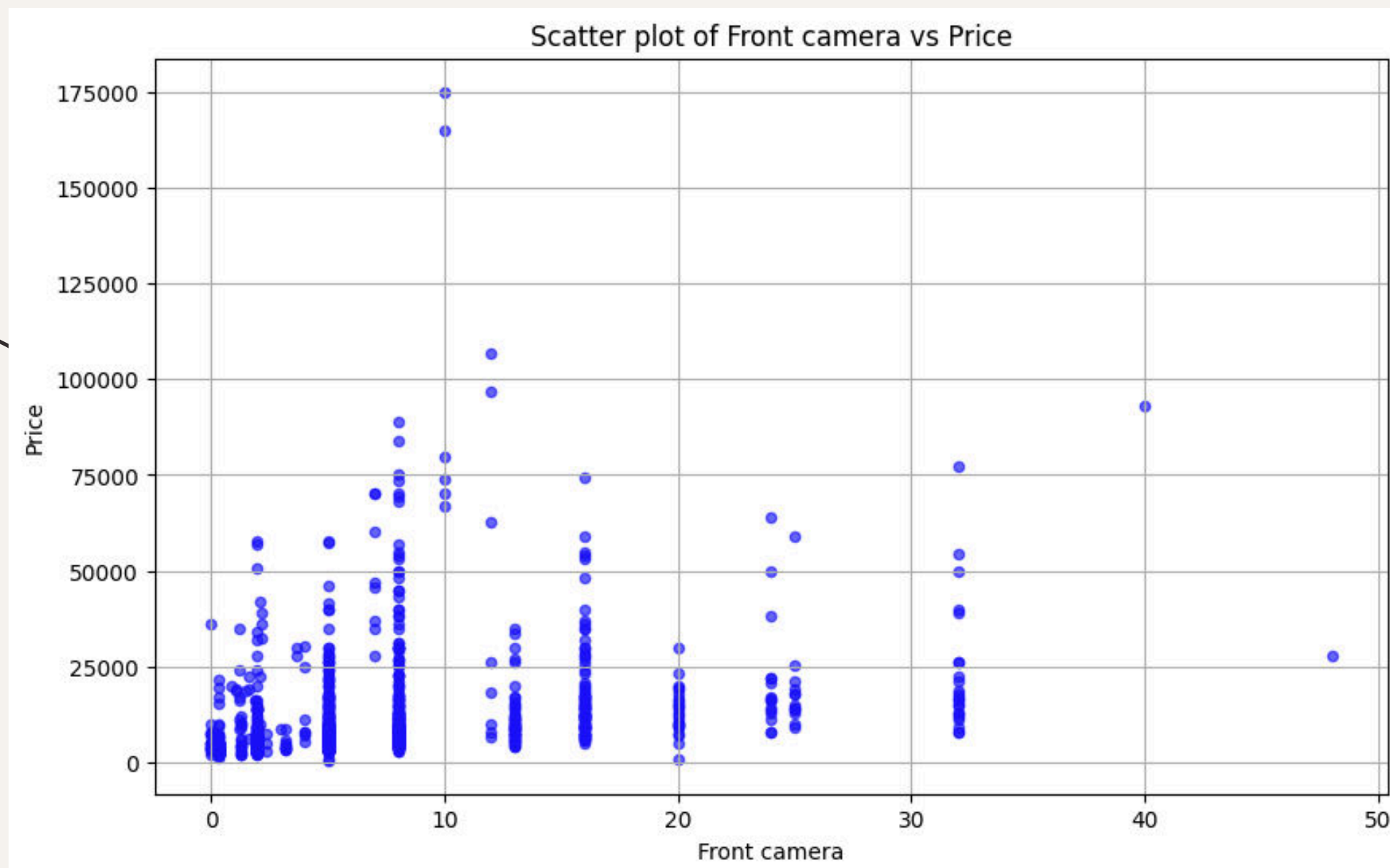
Distribution of prices reveals a skewed pattern, with most prices concentrated in the lower range. This indicates a potential outlier or pricing bias in the dataset, which could impact model performance.

FRONT CAMERA AND PRICE

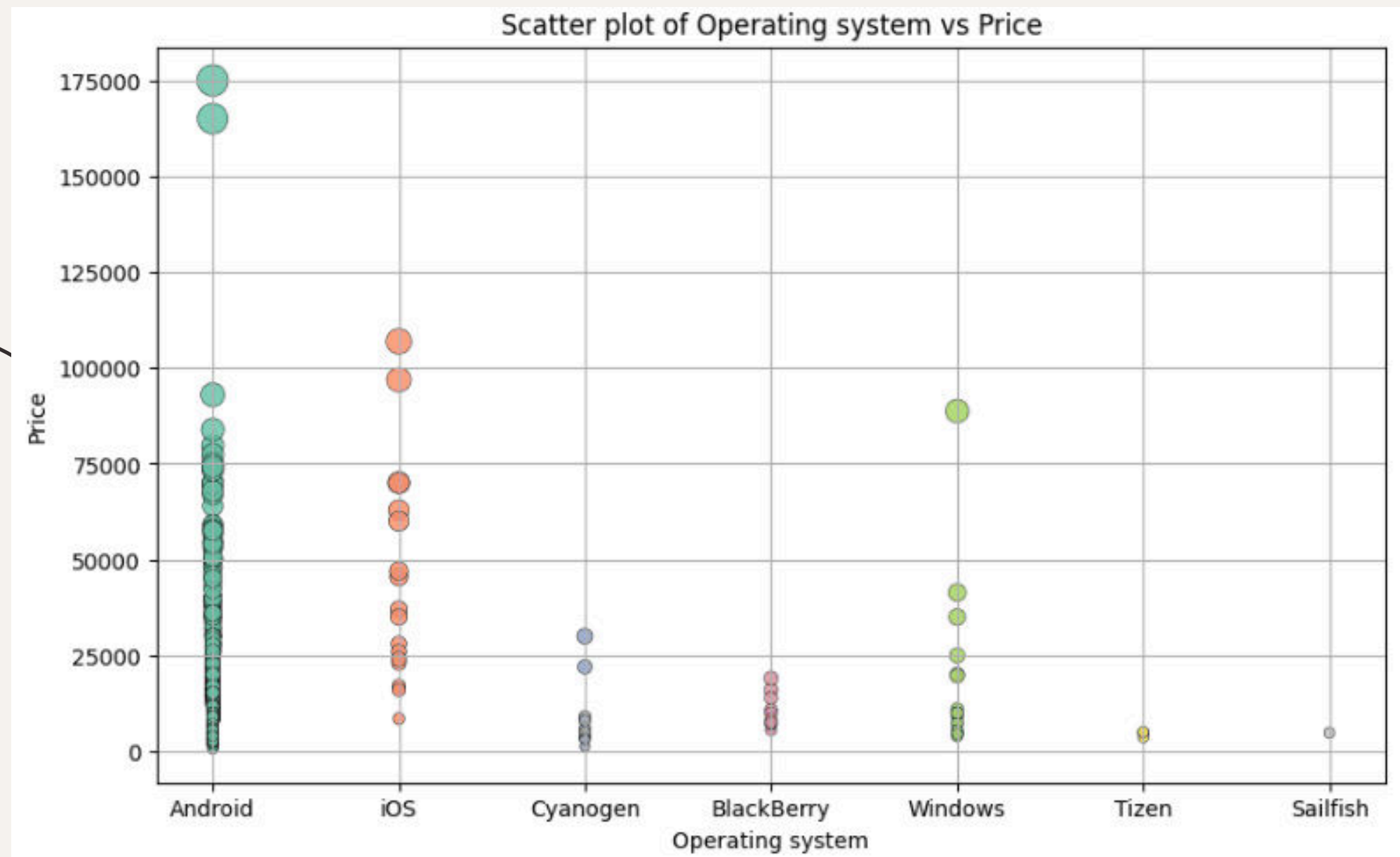
The analysis of the front camera vs price ratio shows a positive correlation, with higher-priced mobile devices typically featuring more advanced front cameras. This suggests that camera quality is a key factor in pricing for higher-end smartphones.



FRONT CAMERA AND PRICE

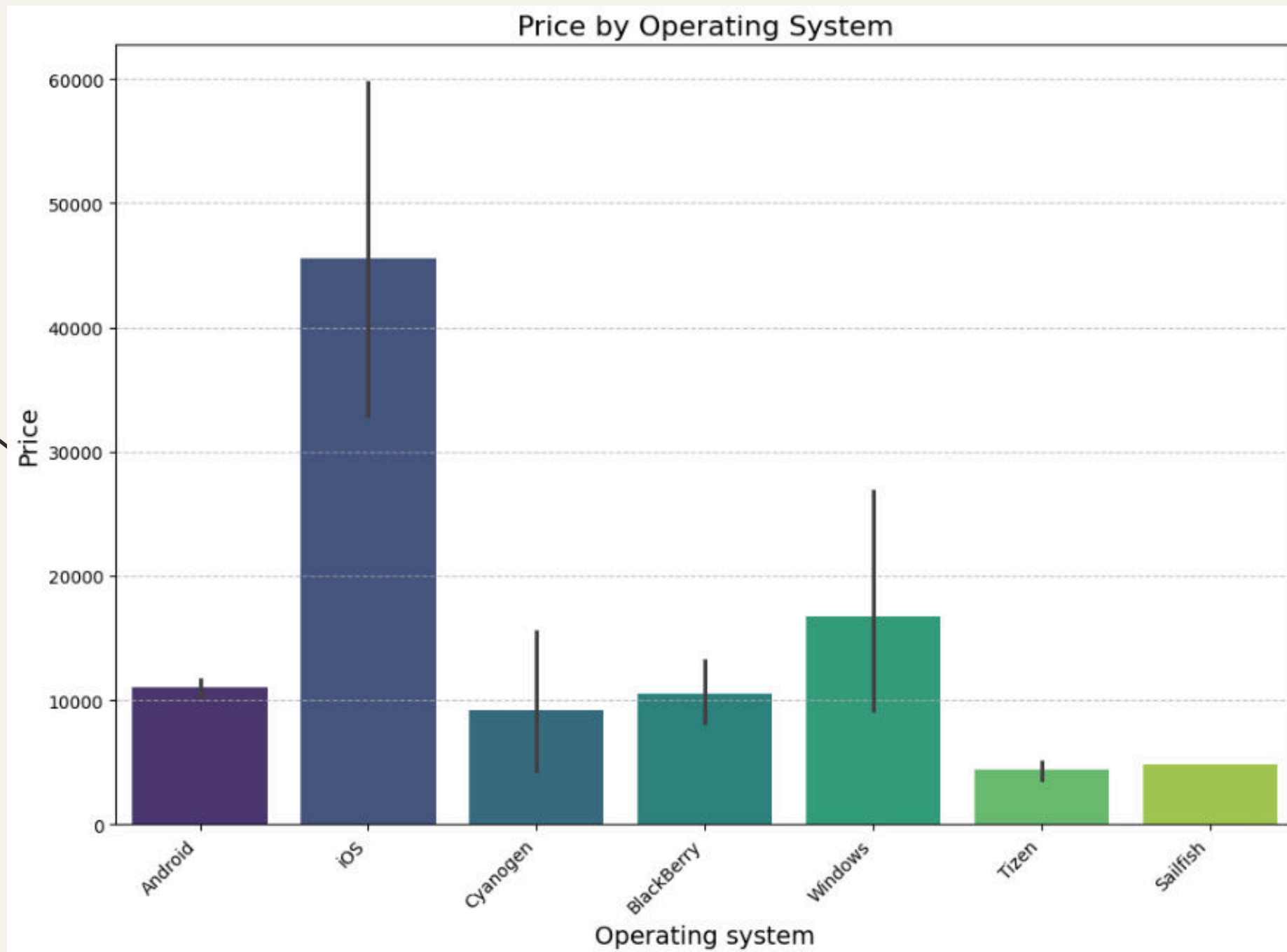


OPERATING SYSTEM TRENDS



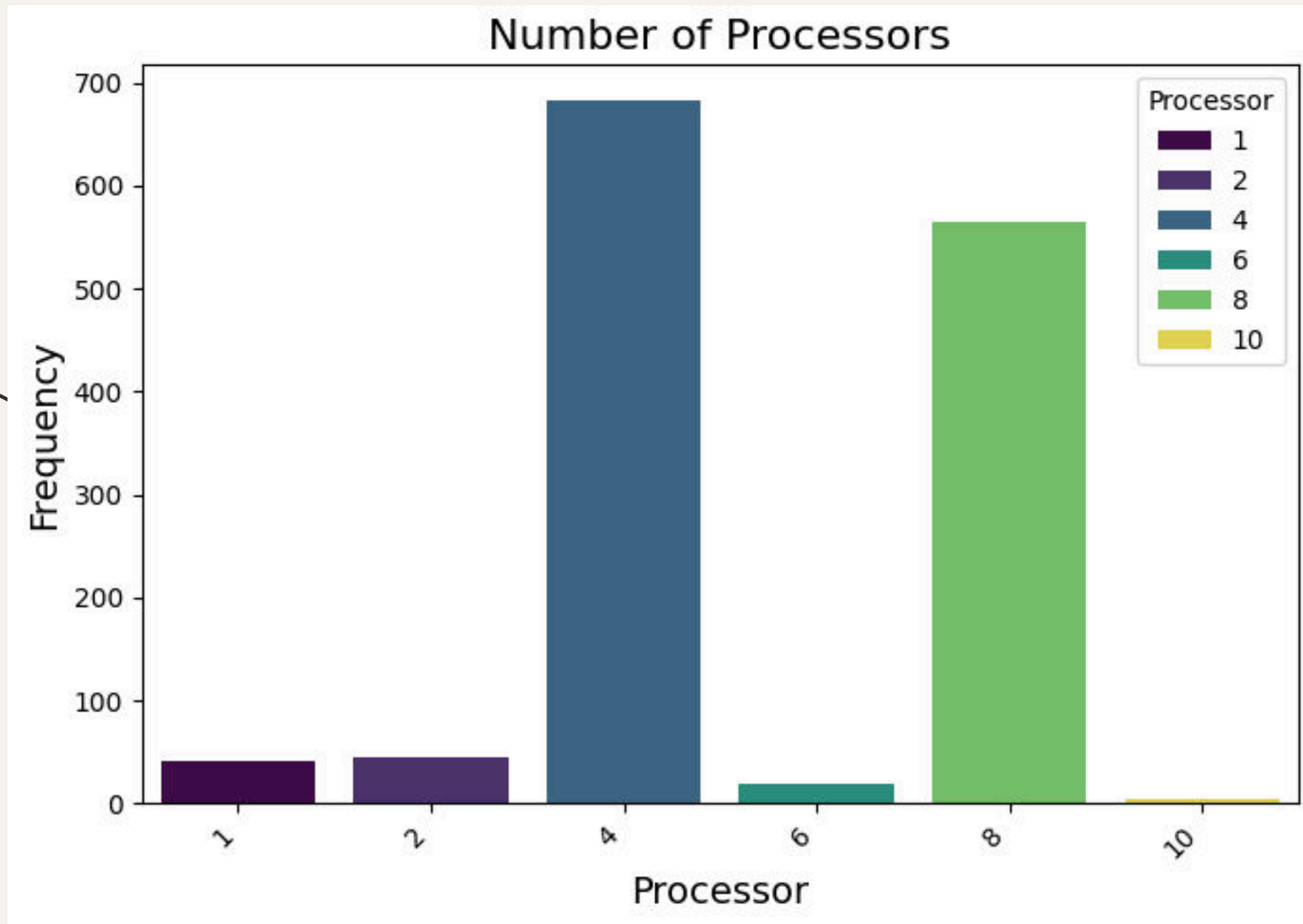
Android leads as the predominant operating system, reflecting a vast and varied market with numerous brands and versions within the Android ecosystem.

PRICE BY OPERATING SYSTEM



Mobile device prices vary by operating system, with iOS devices generally priced higher due to premium features, while Android offers a wider range of price points.

NUMBER OF PROCESSOR



The number of processors in mobile devices varies depending on the model, with high-end smartphones typically featuring multi-core processors (e.g., 6, 8, or more cores) for better performance, while budget devices may have fewer cores.

Data Preprocessing Techniques

Before modeling, data must undergo **preprocessing** ensure accuracy and consistency. We will cover essential techniques like **Lable Encoding** and **OneHotEncoding** for **Catagorical values** to prepare your data for effective analysis and model training.

```
le = LabelEncoder()

for column in ['Brand', 'Model', 'Operating system']:
    le = LabelEncoder()
    x[column] = le.fit_transform(x[column])
```

```
ct = ColumnTransformer(
    transformers=[('encoder', OneHotEncoder(), [4, 13, 14, 15, 17, 18])],
    remainder='passthrough'
)

x = np.array(ct.fit_transform(x))
```

Model Building

Traning the model

```
reg = LinearRegression() # In
reg.fit(X_train, y_train) # F
```

LinearRegression

LinearRegression()

Develop a predictive model for Mobile Price using **Linear Regression**.

Although **KNN** was initially tested, it did not yield satisfactory accuracy. Linear Regression was chosen for its efficiency in handling binary outcomes and its ability to provide clear probability estimates, leading to improved model performance in this context.

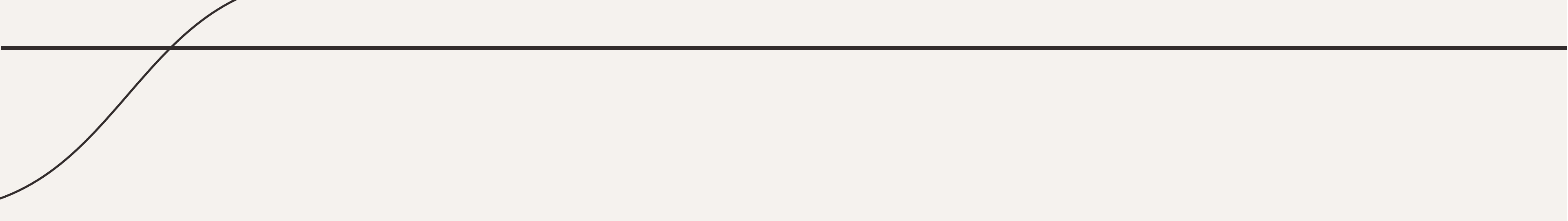
Model Evaluation

```
print(f"Root Mean Squared Error (RMSE): {RMSE:.4f}")  
print(f"Mean Absolute Error (MAE): {MAE:.4f}")  
print(f"R-squared (R2): {R_squared:.4f}")
```

```
Root Mean Squared Error (RMSE): 7653.8744  
Mean Absolute Error (MAE): 4855.3223  
R-squared (R2): 0.7181
```

Conclusion

The model evaluation reveals a Root Mean Squared Error (RMSE) of 7653.8744 and a Mean Absolute Error (MAE) of 4855.3223, suggesting that while there is some prediction error, the model is reasonably accurate. The R-squared (R^2) value of 0.7181 indicates that approximately **72%** of the variance in the data is explained by the model, demonstrating a good fit and effective predictive performance. Overall, the model provides a solid basis for predictions, though there is room for improvement.



Thanks!